

Applied Statistical Modeling 2

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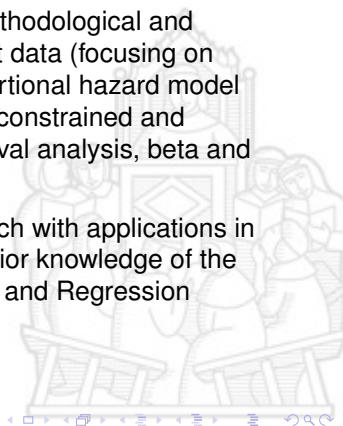


Course description and practical information

Overview

The course aims to provide students with a methodological and applied background of models for time-to-event data (focusing on survival analysis and specifically on Cox proportional hazard model and on models for competing risk), models for constrained and censored response variables focusing on survival analysis, beta and tobit regression

The course foresees a practice-oriented approach with applications in the context of social sciences, and assumes prior knowledge of the foundations of Probability, Inferential Statistics, and Regression models

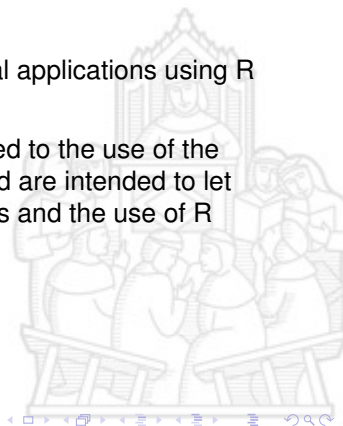


Course description and practical information

Classes

The course foresees both lectures and practical applications using R software.

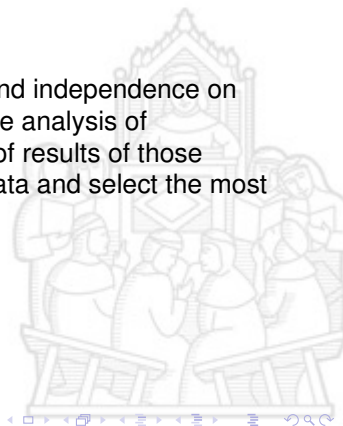
Practical applications deal with examples related to the use of the different models on open source databases and are intended to let students acquire familiarity with the approaches and the use of R software to perform and interpret analyses



Course description and practical information

Goals

Students are expected to acquire knowledge and independence on the use of the most common approaches for the analysis of time-to-event data and on the comprehension of results of those analyses being able to critically comment on data and select the most appropriate approaches for the analysis



Course description and practical information

Final evaluation

The final evaluation consists of a presentation discussing an assignment related to the analysis of an example dataset

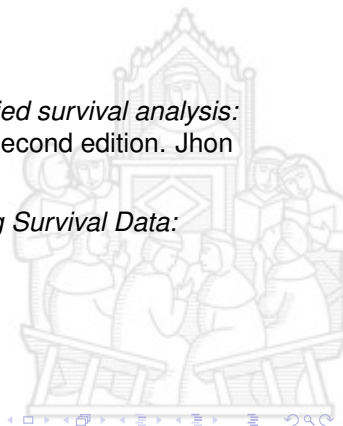


Additional material to deepen models for time-to-event data

Textbooks

Hosmer DW, Lemeshow S, May S (2008). *Applied survival analysis: Regression Modeling of Time-to-Event Data*. Second edition. Jhon Wiley and Sons, Inc., Hoboken, New Jersey

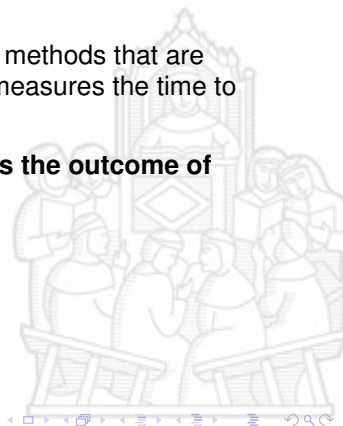
Therneau TM, Grambsch PM (2000). *Modeling Survival Data: Extending the Cox Model*. Springer



Introduction

Survival analysis deals with all those statistical methods that are suited for modeling a dependent variable that measures the time to the occurrence of an event of interest

Not the event per se BUT time to the event is the outcome of interest in time-to-event analysis



What are time-to-event data?

Consider some possible events of interest:

- ▶ First occupation
- ▶ First child
- ▶ Business failure

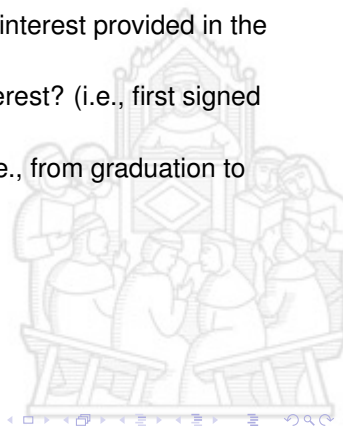
that could be evaluated through a longitudinal study observing a population of interest over time



What are time-to-event data?

Considering the example of possible events of interest provided in the previous slide

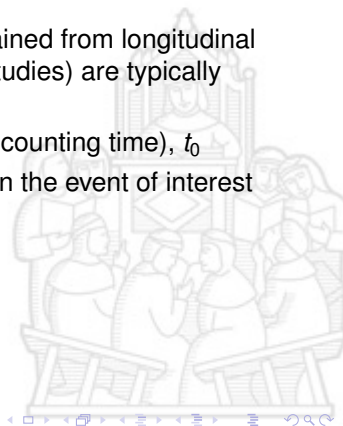
- ▶ How do we define the specific event of interest? (i.e., first signed contract, first delivery, etc.)
- ▶ How do we measure time to the event? (i.e., from graduation to first contract, age, etc.)
- ▶ ...



What are time-to-event data?

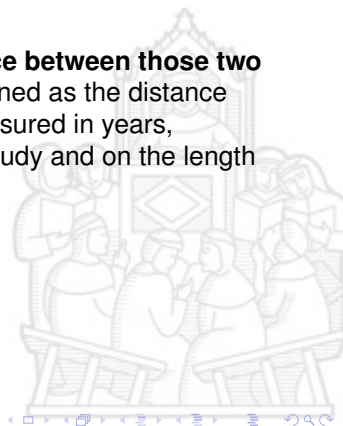
Time-to-event data (survival data) that are obtained from longitudinal studies (either observational or experimental studies) are typically characterized by

- ▶ a beginning (time) point (the time we start counting time), t_0
- ▶ an ending (time) point that is reached when the event of interest occurs, t_1



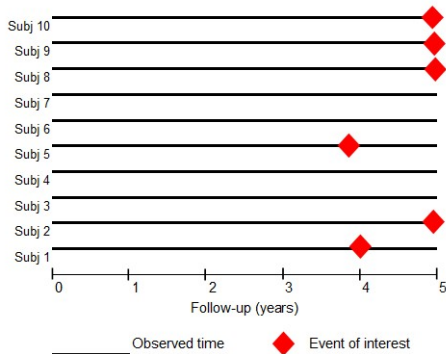
What are time-to-event data?

Thus, **survival time is defined as the distance between those two time points**, which in practice is typically obtained as the distance between two calendar dates and could be measured in years, months, days, etc., depending on the type of study and on the length of the observation (i.e., follow-up time)



Characteristics of time-to-event data

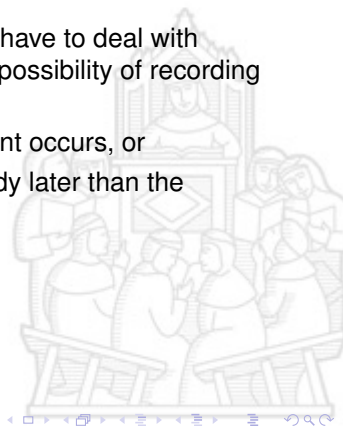
"Ideal" survival data,



Characteristics of time-to-event data

When doing a longitudinal study we frequently have to deal with incomplete data, which could arise from the impossibility of recording the event of interest on some subjects,

- ▶ because they exit the study before the event occurs, or
- ▶ because some subjects may enter the study later than the pre-defined starting date set



Characteristics of time-to-event data

Those *incomplete observations* are typical of survival data and are referred as

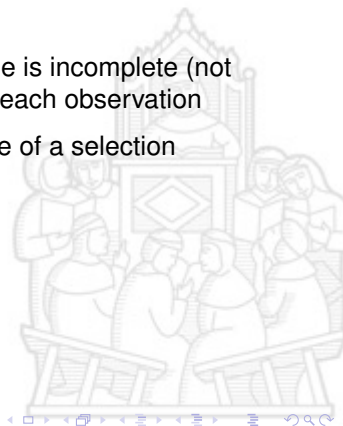
- ▶ **censoring**, and
- ▶ **truncation**



Characteristics of time-to-event data

A **censored** observation is the one whose value is incomplete (not known) because of factors that are random for each observation

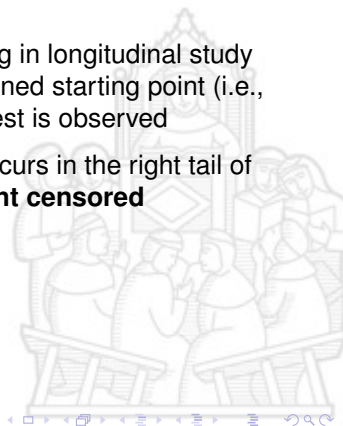
A **truncated** observation is incomplete because of a selection process that is inherent in the study design



Characteristics of time-to-event data

The most common form of **censoring** occurring in longitudinal study is the one where observation begins at the defined starting point (i.e., $t = 0$) and terminates before the event of interest is observed

As the incomplete nature of the observation occurs in the right tail of the time axis, that observation is said to be **right censored**



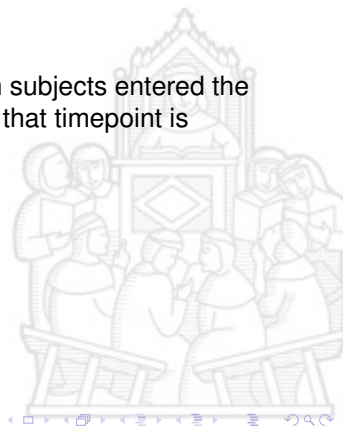
Characteristics of time-to-event data

Another form of censoring that sometimes happens when conducting longitudinal studies is **left censored** which occurs when the event of interest has already occurred when the observation begins

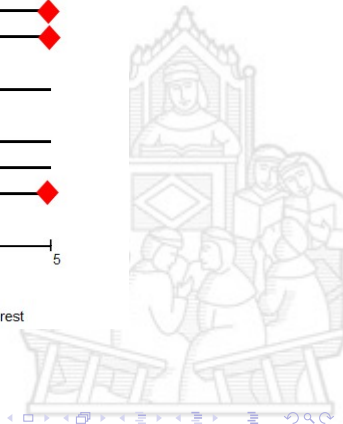
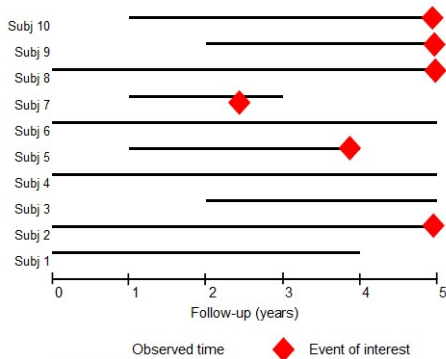


Characteristics of time-to-event data

Left truncation or **delayed entry** occurs when subjects entered the study not at the beginning time points but after that timepoint is already passed

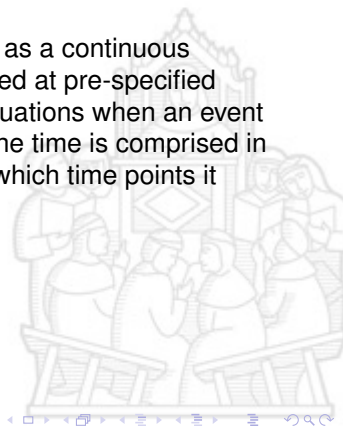


Characteristics of time-to-event data



Characteristics of time-to-event data

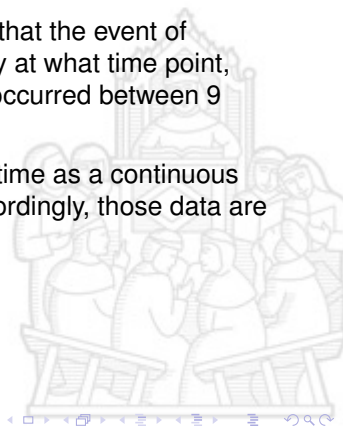
In most studies, time is collected and recorded as a continuous variable, while sometimes subjects are contacted at pre-specified intervals (i.e., every 3 months), and in those situations when an event occurs at a certain interval, we just know that the time is comprised in a certain interval but we don't know exactly at which time points it happens



Characteristics of time-to-event data

For example, if a subject reports at 12 months that the event of interest has occurred but we don't know exactly at what time point, the only information we have is that the event occurred between 9 and 12 months

In those cases, it is not recommended to treat time as a continuous variable, and to use models that treat time accordingly, those data are said to be **interval censored**



Describing time-to-event data

Because of the characteristics of time-to-event data, the description of those data must begin with the estimation of the cumulative distribution function

Remember that, given a random variable X , the cumulative distribution function is defined as

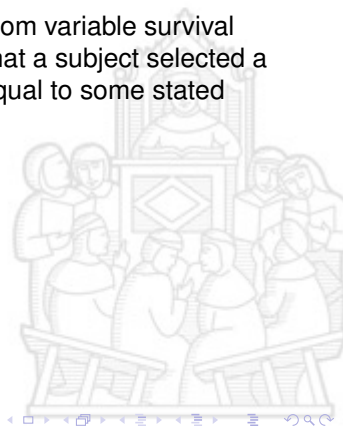
$$F(x) = P(X \leq x^*)$$



Describing time-to-event data

The cumulative distribution function of the random variable survival time, denoted T , is defined as the probability that a subject selected at random will have a survival time less than or equal to some stated value, t

$$F(t) = P(T \leq t)$$



Describing time-to-event data

The survival is thus the probability of observing a survival time greater than some stated value, t

$$S(t) = P(T > t)$$

and by definition

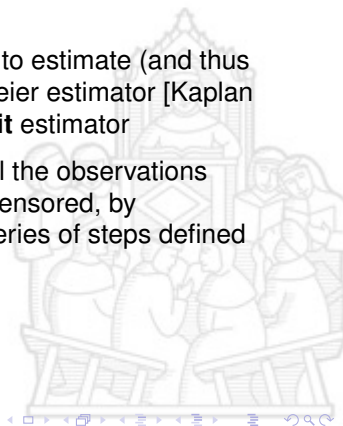
$$S(t) = 1 - F(t)$$



Describing time-to-event data (Kaplan-Meier estimator)

One of the most widespread approaches used to estimate (and thus describe) the survival function is the Kaplan-Meier estimator [Kaplan and Meier (1958)], also called the **product limit** estimator

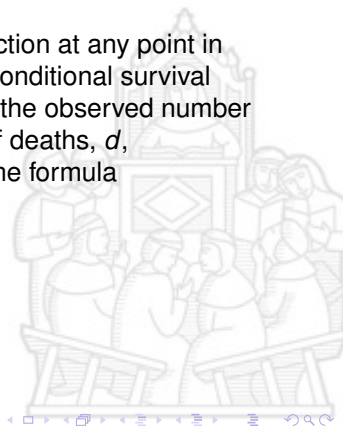
This estimator incorporates information from all the observations available, both uncensored (event times) and censored, by considering survival to any point in time as a series of steps defined at the observed survival and censored times



Describing time-to-event data (Kaplan-Meier estimator)

The Kaplan-Meier estimator of the survival function at any point in time is obtained by multiplying a sequence of conditional survival probability estimators which are obtained from the observed number at risk of death, n , and the observed number of deaths, d , (conditional on given time point) according to the formula

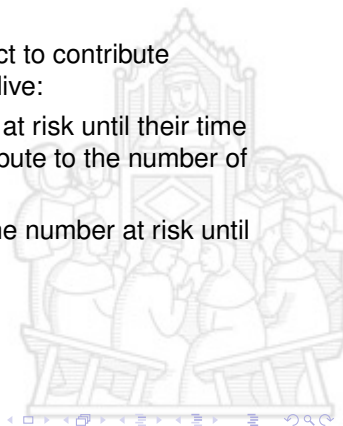
$$\frac{(n - d)}{n}$$



Describing time-to-event data (Kaplan-Meier estimator)

The Kaplan-Meier estimator allows each subject to contribute information as long as he/she is known to be alive:

- ▶ subjects who die contribute to the number at risk until their time of death, and at this point, they also contribute to the number of deaths
- ▶ subjects who are censored contribute to the number at risk until they are lost to follow-up

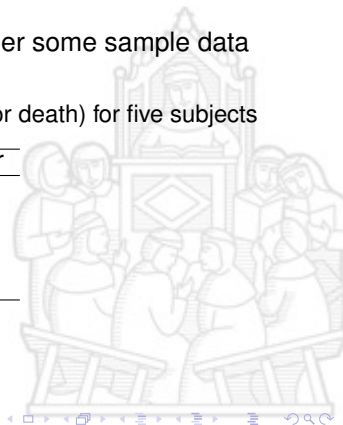


Describing time-to-event data (Kaplan-Meier estimator)

To illustrate the Kaplan-Meier estimator, consider some sample data

Table: Survival times and vital status (Censor==1 for death) for five subjects

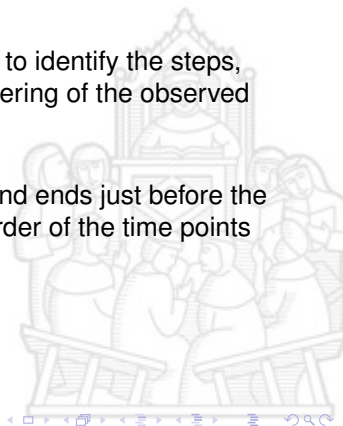
Subject	Time	Censor
1	6	1
2	44	1
3	21	0
4	14	1
5	62	1



Describing time-to-event data (Kaplan-Meier estimator)

To derive the Kaplan-Meier estimator, we need to identify the steps, that consist of intervals defined by the rank ordering of the observed time

i.e., each interval begins at an observed time and ends just before the next ordered time and is indexed by the rank order of the time points defining its beginning



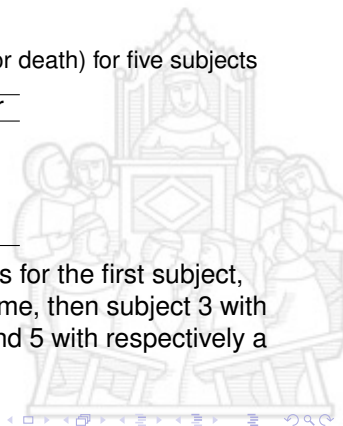
Describing time-to-event data (Kaplan-Meier estimator)

Using the example data shown before

Table: Survival times and vital status (Censor==1 for death) for five subjects

Subject	Time	Censor
1	6	1
2	44	1
3	21	0
4	14	1
5	62	1

the first rank-ordered of observed time is 6 days for the first subject, followed by subject 4 with 14 days of survival time, then subject 3 with 21 days of survival time, and then subjects 2 and 5 with respectively a survival time of 44 and 62 days



Describing time-to-event data (Kaplan-Meier estimator)

The intervals of rank-ordered time that could be identified are thus

$$I_0 = t : 0 \leq t < 6 = [0, 6)$$

$$I_1 = t : 6 \leq t < 14 = [6, 14)$$

$$I_2 = t : 14 \leq t < 21 = [14, 21)$$

$$I_3 = t : 21 \leq t < 44 = [21, 44)$$

$$I_4 = t : 44 \leq t < 62 = [44, 62)$$

$$I_5 = t : t \geq 62 = [62, \infty)$$



Describing time-to-event data (Kaplan-Meier estimator)

For each interval defined, it is possible to estimate the **conditional probability** of surviving (conditional on those who survived to the interval under consideration) based on the number of subjects at risk n_i and the number of death d_i in each interval as

$$\frac{(n_i - d_i)}{n_i}$$



Describing time-to-event data (Kaplan-Meier estimator)

The Kaplan-Meier estimator of the survival function is defined the cumulative product of the conditional probability over all steps

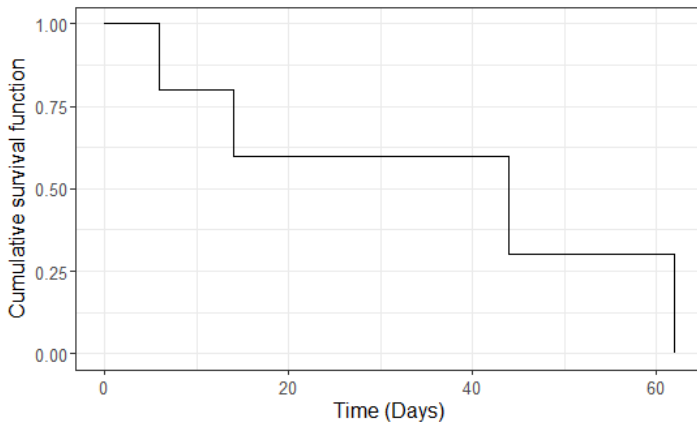
$$\hat{S} = \prod_i \frac{(n_i - d_i)}{n_i}$$



Table: Estimated survival function estimated from Survival times and vital status (Censor==1 for death) for five subjects

Interval	Number at risk	Death	$\frac{(n-d)}{n}$	$\hat{S}$
[0, 6)	6	0	$(6 - 0)/6 = 1$	1.0
[6, 14)	5	1	$(5 - 1)/5 = 0.8$	0.8
[14, 21)	4	1	$(4 - 1)/4 = 0.75$	0.6
[21, 44)	3	0	$(3 - 0)/3 = 1$	0.6
[44, 62)	2	1	$(2 - 1)/2 = 0.5$	0.3
[62, ∞)	1	1	$(1 - 1)/0 = 0$	0

Describing time-to-event data (Kaplan-Meier estimator)

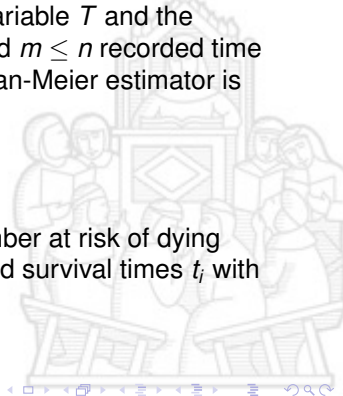


Describing time-to-event data (Kaplan-Meier estimator)

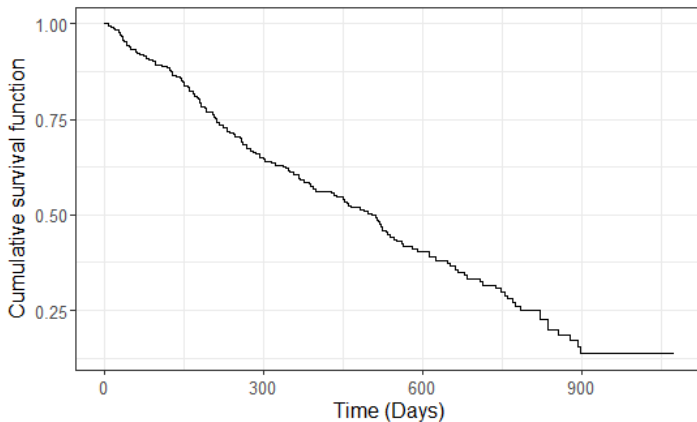
In general, assume to have a sample of n independent observation (t_i, c_i) , $i = 1, 2, \dots, n$ of the underlying survival variable T and the censoring indicator variable C , having observed $m \leq n$ recorded time of failure and $n - m$ censored values, the Kaplan-Meier estimator is given by

$$\hat{S}(t) = \prod_{t_{(i)} \leq t} \frac{(n_i - d_i)}{n_i}$$

where n_i and d_i represent respectively the number at risk of dying and the number of deaths over the rank-ordered survival times t_i with $t_1 < t_2 < \dots < t_m$



Describing time-to-event data (Kaplan-Meier estimator)

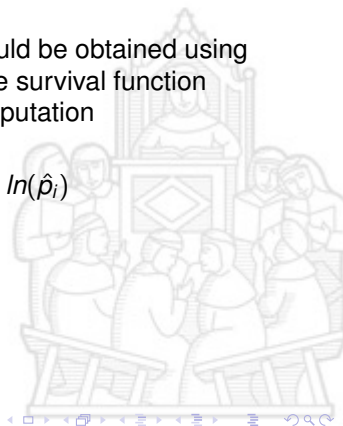


Describing time-to-event data (Kaplan-Meier estimator)

The variance of the Kaplan-Meier estimator could be obtained using the **delta method** and referring to the log of the survival function instead of the function itself to simplify the computation

$$\ln(\hat{S}(t)) = \sum_{t_{(i)} \leq t} \ln\left(\frac{n_i - d_i}{n_i}\right) = \ln(\hat{p}_i)$$

where $\hat{p}_i = (n_i - d_i)/n_i$

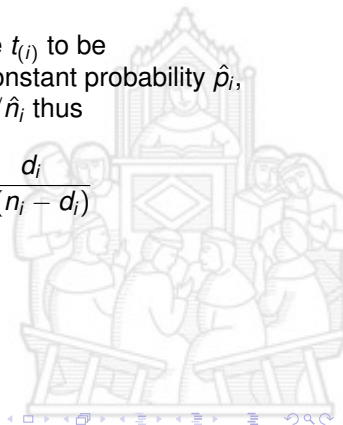


Describing time-to-event data (Kaplan-Meier estimator)

Considering observations in the risk set at time $t_{(i)}$ to be independently distributed as a Bernuolli with constant probability $\hat{p}_i$, its variance could be estimated as $(\hat{p}_i(1 - \hat{p}_i))/\hat{n}_i$ thus

$$\hat{Var}[\ln(\hat{p}_i)] \approx \frac{1}{\hat{p}_i^2} \frac{\hat{p}_i(1 - \hat{p}_i)}{n_i} \approx \frac{d_i}{n_i(n_i - d_i)}$$

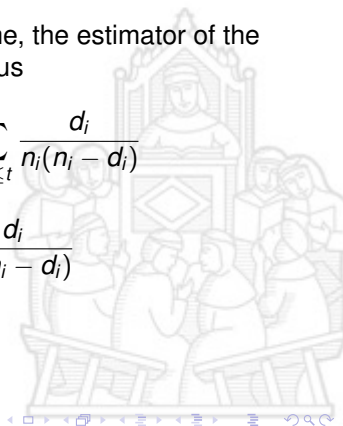
as $Var[\ln(X)] \approx \frac{1}{\mu_x^2} \sigma_x^2$



Describing time-to-event data (Kaplan-Meier estimator)

Assuming independent observation at each time, the estimator of the variance of the log of the survival function is thus

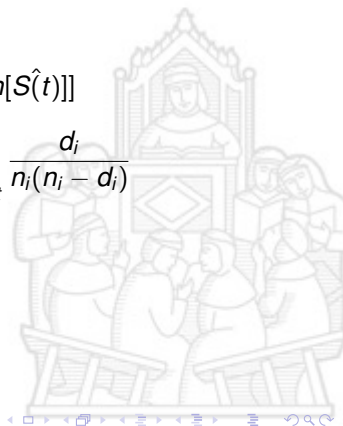
$$\hat{Var}[\ln(\hat{S}(t))] = \sum_{t_{(i)} \leq t} \hat{Var}[\ln(\hat{p}_i)] = \sum_{t_{(i)} \leq t} \frac{d_i}{n_i(n_i - d_i)}$$
$$\Rightarrow \hat{Var}[\hat{S}(t)] = \hat{S}(t)^2 \sum_{t_{(i)} \leq t} \frac{d_i}{n_i(n_i - d_i)}$$



Describing time-to-event data (Kaplan-Meier estimator)

Similarly for the *log-log survival function*, $\ln[-\ln(\hat{S}(t))]$

$$\hat{Var}[\ln[-\ln(\hat{S}(t))]] = \frac{1}{[\ln(\hat{S}(t))]^2} \sum_{t_{(i)} \leq t} \frac{d_i}{n_i(n_i - d_i)}$$

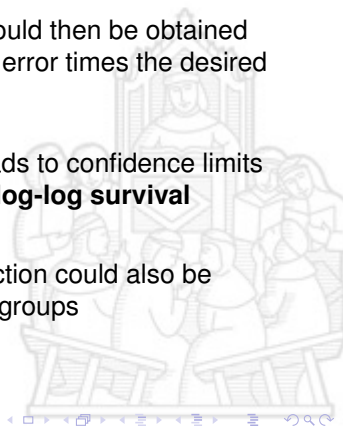


Describing time-to-event data (Kaplan-Meier estimator)

Confidence intervals for the survival function could then be obtained by using the product of the estimated standard error times the desired quantile of the standard normal distribution

As the use of the **delta method** sometimes leads to confidence limits below zero or greater than one, the use of the **log-log survival function** is preferred to address this problem

Point and interval estimates of the survival function could also be used to compare survival experience between groups



Describing time-to-event data (Kaplan-Meier estimator)

The general idea behind the calculation of each test is based on a contingency table of groups by status at each observed survival time

Table: Table used for test of equality of the survival function in two groups at observed survival time t_i

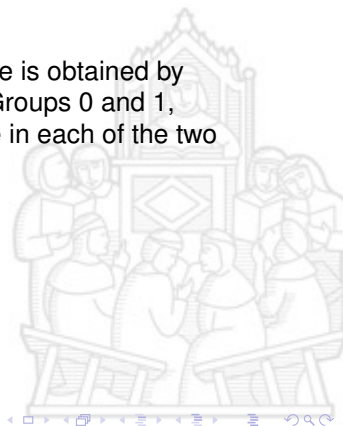
Event	Group 1	Group 0	Total
Die	d_{1i}	d_{0i}	d_i
Not die	$n_{1i} - d_{1i}$	$n_{0i} - d_{0i}$	$n_i - d_i$
At risk	n_{1i}	n_{0i}	n_i

where n_{ji} and d_{ji} , $j = 0, 1$ are respectively the number at risk and observed deaths in Group 0 and 1 at time t_i

Describing time-to-event data (Kaplan-Meier estimator)

The contribution to the test statistic at each time is obtained by calculating the expected number of deaths in Groups 0 and 1, assuming that the survival function is the same in each of the two groups, that is given by

$$\hat{e}_{ji} = \frac{n_{ji}d_i}{n_i}$$

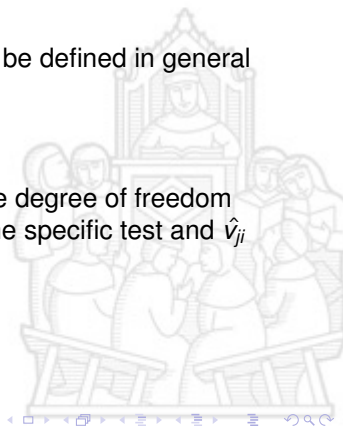


Describing time-to-event data (Kaplan-Meier estimator)

The tests for comparison between groups may be defined in general as

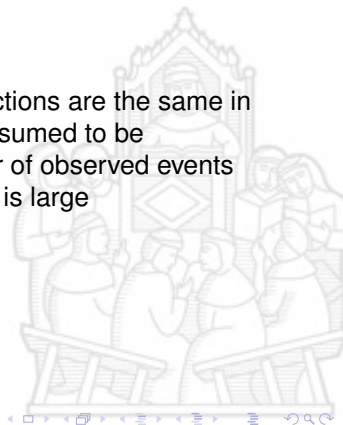
$$Q = \frac{[\sum_{i=1} w_i (d_{ji} - \hat{e}_{ji})]}{\sum_{i=1} w_i^2 \hat{v}_{ji}}$$

which follows a Chi-square distribution with one degree of freedom and w_i are weights whose values depend on the specific test and $\hat{v}_{ji}$ is the estimator of the variance of d_{ji}



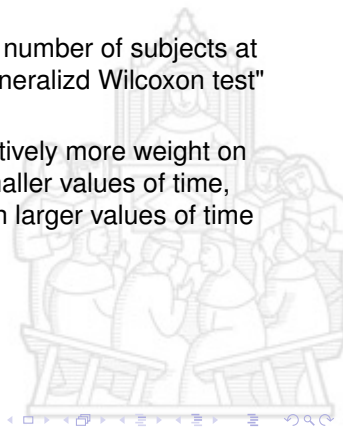
Describing time-to-event data (Kaplan-Meier estimator)

Under the null hypothesis, the two survival functions are the same in the two groups, the censoring experience is assumed to be independent of the group, and the total number of observed events and the sum of the expected number of events is large



- ▶ The well-known Log-rank test, due to Peto and Peto (1972) consider $w_i = 1$
- ▶ When $w_i = n_i$, so weights are equal to the number of subjects at risk at each survival time, we have the "generalized Wilcoxon test" due Gehan (1965) and Breslow (1970)

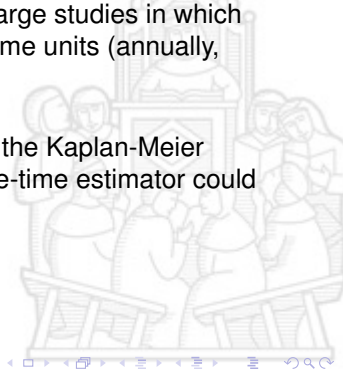
While the "generalized Wilcoxon test" puts relatively more weight on differences between the survival function at smaller values of time, the Log-rank test puts relatively more weight on larger values of time



Describing time-to-event data (Life-time estimator)

A grouped-data analog of the Kaplan-Meier estimator is the life-time estimator that has been used over a long time to describe mortality data and well-fit situations when we deal with large studies in which mortality experience is presented in calendar time units (annually, semi-annually, etc.)

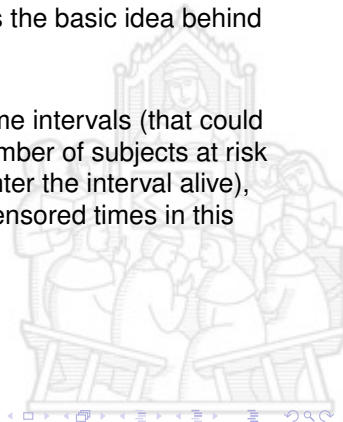
In these situations, the tabulation and graph of the Kaplan-Meier estimator can be quite cumbersome and the life-time estimator could be preferable



Describing time-to-event data (Life-time estimator)

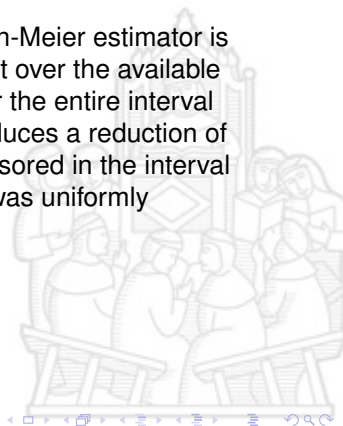
The derivation of the life-table estimator follows the basic idea behind the Kaplan-Meier estimator

Assume to having observed data over some time intervals (that could be year or semester) and let again n be the number of subjects at risk of dying at time t (also said the number who enter the interval alive), and d and c be subjects having survival and censored times in this interval



Describing time-to-event data (Life-time estimator)

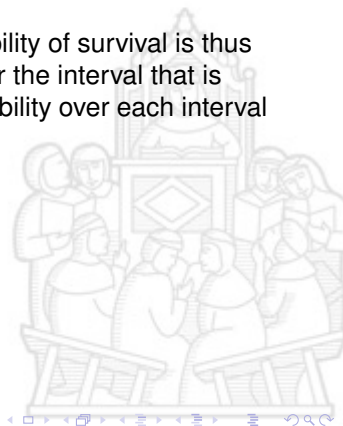
What differentiates the life-time from the Kaplan-Meier estimator is that the first one takes into account the fact that over the available intervals, not all subjects are at risk of dying for the entire interval. To account for that the life-time estimator introduces a reduction of the size of the risk set by one-half of those censored in the interval, thus assuming that the censored observation was uniformly distributed over the interval.



Describing time-to-event data (Life-time estimator)

In the life-time estimator the conditional probability of survival is thus estimated considering the average risk set over the interval that is $n - (c - 2)$, it follows that the conditional probability over each interval i is derived as

$$\frac{(n_i - (c_i/2) - d_i)}{n_i - (c_i/2)}$$



Describing time-to-event data (Life-time estimator)

Using the example data shown before

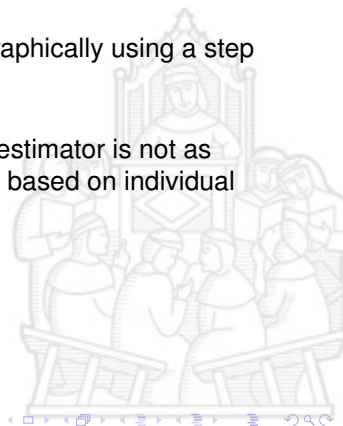
Table: Survival times and vital status (Censor==1 for death) for five subjects

Interval	Enter	Die	Censored	S
[0, 1)	100	20	0	$\frac{(100 - (0/2) - 20)}{(100 - (0/2))} = 0.80$
[1, 2)	80	5	0	0.75
[2, 3)	75	7	0	0.68
[3, 4)	68	4	0	0.64
[4, 5)	64	6	0	0.58
[5, 6)	58	58	38	0.51
[6, 7)	15	2	1	0.44
[7, 8)	12	2	10	0.31

Describing time-to-event data (Life-time estimator)

The life-time estimator could be represented graphically using a step function or a polygon representation

Being derived from grouped data, the life-time estimator is not as precise as the Kaplan-Meier estimator which is based on individual data



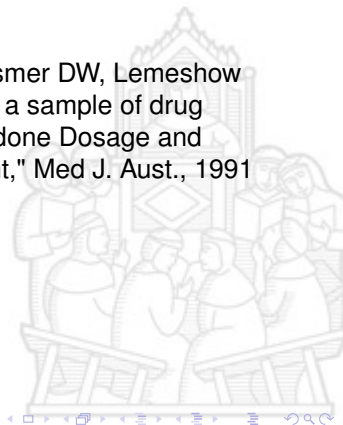
Example dataset

For both the practical application and the assignment, we will use datasets freely available online



Addicts dataset

The "addicts" dataset is also considered in Hosmer DW, Lemeshow S, May S ' book and consists of data related to a sample of drug users from a study by Caplehorn et al. "Methadone Dosage and Retention of Patients in Maintenance Treatment," Med J. Aust., 1991



Addicts dataset

The dataset contains data on 238 subjects identified by an anonymous code, and the Stata format can be downloaded from:

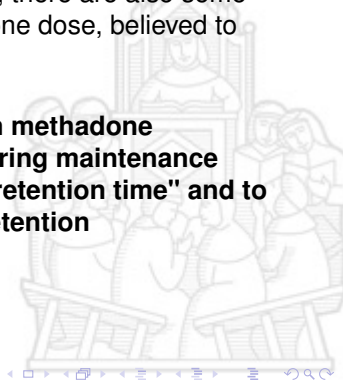
<http://web1.sph.emory.edu/dkleinb/allDatasets/surv2datasets/addicts.dta>



Addicts dataset

Data comprise time (in days) spent by heroin addicts from entry to departure from two different methadone clinics, there are also some covariates, namely, prison record and methadone dose, believed to affect the survival time

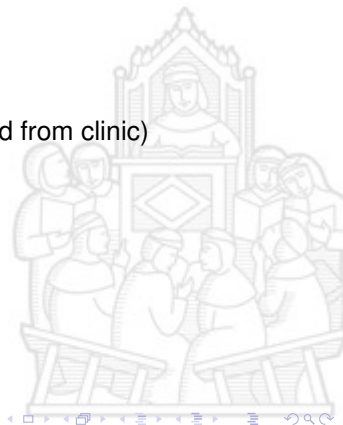
We will use these data to evaluate **retention in methadone treatment in a cohort of heroin addicts entering maintenance programmes to evaluate both understand "retention time" and to evaluate potential variables impacting on retention**



Addicts dataset

List of variables

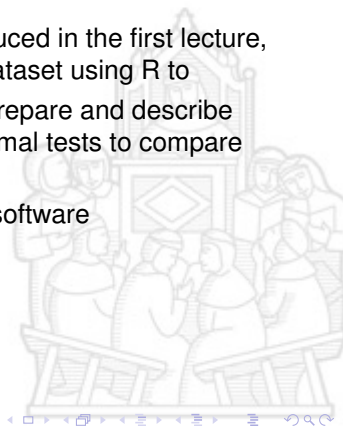
- ▶ Subject ID
- ▶ Clinic (1 or 2)
- ▶ Survival status (0 = censored, 1 = departed from clinic)
- ▶ Survival time in days
- ▶ Prison record (0 = none, 1 = any)
- ▶ Methadone dose (mg/day)



Analysis of survival data

Based on the concepts and approaches introduced in the first lecture, we will have a simple analysis of the *addicts* dataset using R to

- ▶ both go into specific examples of how to prepare and describe survival data, interpret results, perform formal tests to compare survival among groups, and
- ▶ learning how to do that in practice with R software



Analysis of Addicts in R

A simple preview of some records of the *addicts* dataset

id Subject ID	clinic Coded 1 or 2	status status (0=censored, 1=endpoint)	survt survival time in days	prison 0=none, 1=prison record	dose methadone dose (mg/day)
1	1	1	428	0	50
2	1	1	275	1	55
3	1	1	262	0	55
4	1	1	183	0	30
5	1	1	259	1	65
6	1	1	714	0	55
7	1	1	438	1	65
8	1	0	796	1	60
9	1	1	892	0	50
10	1	1	393	1	65

Analysis of Addicts data in R

There are several packages in R that could be used to perform survival analysis and graph survival curves, and we will see some of them, for today's class, we will use `survival` and `ggsurfit`



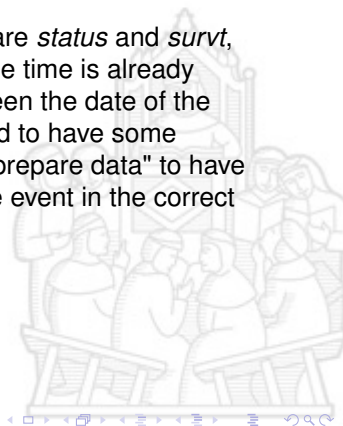
Analysis of Addicts data in R

As we discussed in the first lecture the most important thing in the analysis of survival data is related to the identification of the variables of interest

id Subject ID	clinic Coded 1 or 2	status status (0=censored, 1=endpoint)	survt survival time in days	prison 0=none, 1=prison record	dose methadone dose (mg/day)
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8	1	0	796	1	60
9	1	1	892	0	50
10	1	1	393	1	65

Analysis of Addicts data in R

!! In this case, the variables of interest, which are *status* and *survt*, are "ready to be used", as for example, because time is already recorded in the dataset as the difference between the date of the event/date of the last follow-up without the need to have some previous calculus, in some cases we need to "prepare data" to have the time variable and the variable related to the event in the correct format to be used from software



Analysis of Addicts data in R

Once variables of interest have been identified and prepared, we need to create an R object containing the variable of interest in order to let R properly "understand" and use survival data, this could be done using the function `Surv` the by typing:

```
Surv(survt,status)
```



Analysis of Addicts data in R

We can have a look on the object created, and for the ease of reading, we can ask R to see just the first twenty entries of the object, typing:

```
Surv(survt,status) [1:20]
```

```
[1] 428 275 262 183 259 714 438 796+ 892 393 161+ 836  
[13] 523 612 212 399 771 514 512 624
```

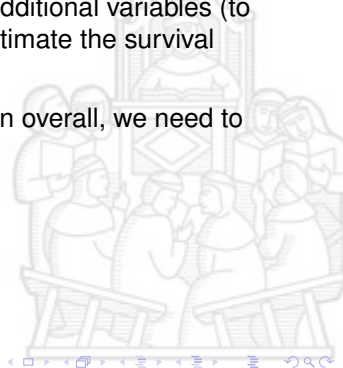
R just created an object of length equal to the number of units in the file, and made up of the survival time for each subject, which is complemented by a *plus* for those subjects experiencing the event of interest, that in our case is treatment failure

Analysis of Addicts data in R

Once we set up the data, it is possible to estimate the survival function using the Kaplan-Meier estimator with the function `survfit`, which requires a survival object and eventual additional variables (to be added after the tilde) to be considered to estimate the survival function within different groups.

As we just want to estimate the survival function overall, we need to add 1 after the tilde:

```
survfit(Surv(survt, status) ~ 1, data=addicts)
```



Analysis of Addicts data in R

The output of the `survfit` function provides the estimated median survival time according to the Kaplan-Meier approach and its 95% confidence interval

```
call: survfit(formula = Surv(addicts$survt, addicts$status) ~ 1, data = addicts)
```

	n	events	median	0.95LCL	0.95UCL
[1,]	238	150	504	399	560

The estimate of median survival time corresponds to the survival time when the survival function is equal to 50%, in our case we estimated a median retention on treatment of about 504 days (less than 1 year and a half) with a 95% confidence interval varying between 399 and 560 days

Analysis of Addicts data in R

By asking a `summary` of the object created with `survfit` we can also obtain estimates of the survival function at a given time point, for example typing:

```
summary(survfit(Surv(survt, status) ~ 1, data=addicts),times=365.25)
```

we obtain an estimate of the survival function and its 95% confidence interval at one year that is about 61% with a 95% confidence interval comprised between about 55% and 68%, at one year a total of 122 are still on treatment while till that time a total of 87 treatment withdrawn were recorded

```
Call: survfit(formula = Surv(addicts$survt, addicts$status) ~ 1, data = addicts)
```

time	n.risk	n.event	survival	std.err	lower 95% CI	upper 95% CI
365	122	87	0.606	0.0331	0.545	0.675

Analysis of Addicts data in R

While typing :

```
summary(survfit(Surv(survt, status) ~ 1,  
data=addicts),times=c(30,60,90,180,365.25))
```

allows obtaining an estimate of the survival function and its 95% confidence interval at different time points that could be specified depending on the specific interest



Analysis of Addicts data in R

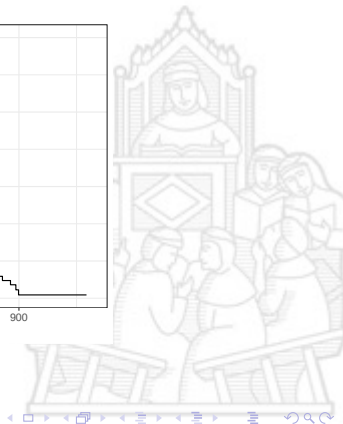
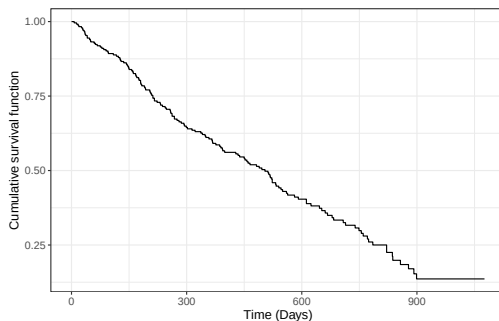
From the example we understand that retention on treatment is 97% (95% CI: 95%-99%) at 1 month, it decreases to about 93% (95% CI: 90%-96%) at 2 months, 90% (95% CI: 86%-94%) at 3 months and drop to about 80% (95% CI: 74%-85%) at 6 months

```
call: survfit(formula = surv(addicts$survt, addicts$status) ~ 1, data = addicts)
```

time	n.risk	n.event	survival	std.err	lower	95% CI upper	95% CI
30	228	7	0.970	0.0111	0.949	0.992	
60	215	10	0.927	0.0169	0.895	0.961	
90	207	6	0.901	0.0195	0.864	0.940	
180	175	24	0.794	0.0269	0.743	0.848	
365	122	40	0.606	0.0331	0.545	0.675	

Analysis of Addicts data in R

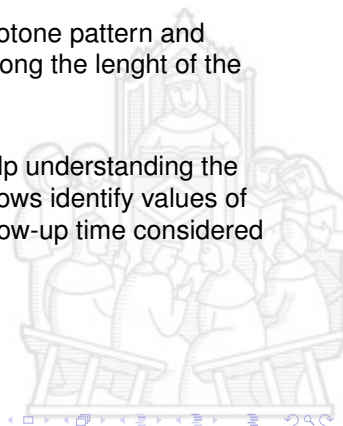
To have a nice plot of the survival curves estimated using the Kaplan-Meier function, we can use the `survfit2`



Analysis of Addicts data in R

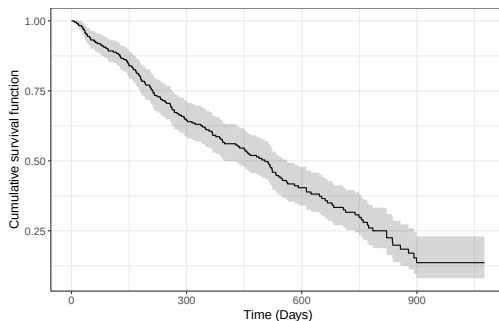
The survival function plotted has a typical monotone pattern and allows identify values of the survival function along the lenght of the follow-up time considered

A visual analysis of the survival curves also help understanding the speed of decrease of the relapse timen and allows identify values of the survival function along the lenght of the follow-up time considered



Analysis of Addicts data in R

It is possible to add 95% confidence interval for the curve just by adding `add_confidence_interval()` to the command used before

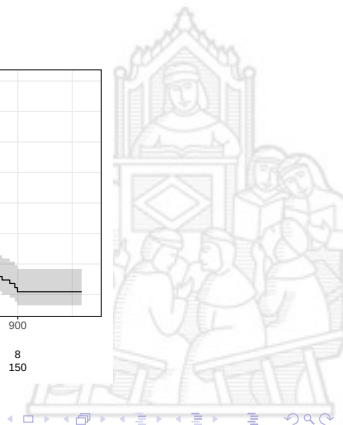
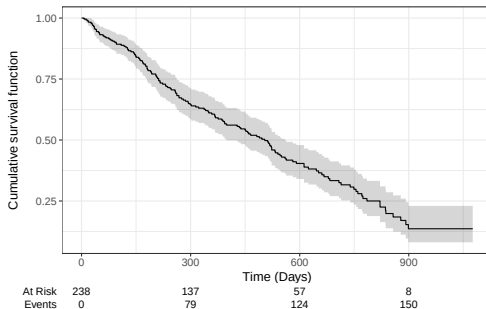


The 95% confidence interval becomes wider as the follow-up time goes on

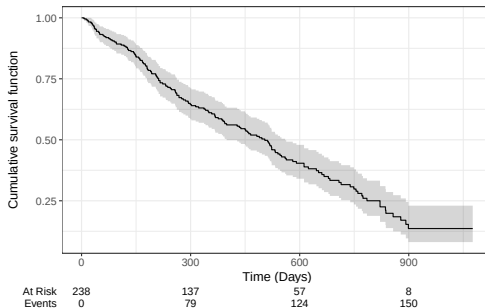


Analysis of Addicts data in R

Adding also `add_risktable()` to the command used before, we can show the risk table below the plot that allows clearly identifying the number of subjects at risk and the number of events that occurred over different time points



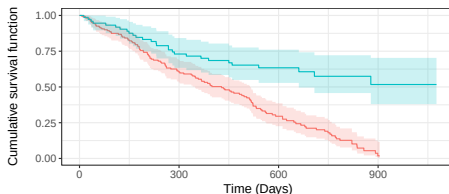
Analysis of Addicts data in R



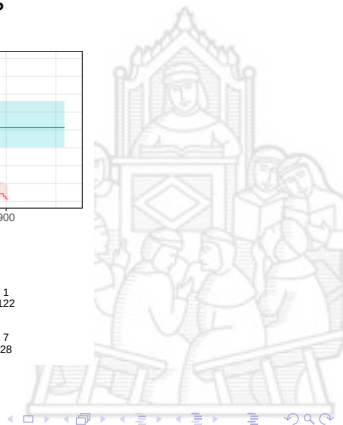
In line with what was discussed in the first lecture, the number of subjects at risk decreased over time, according to both subjects "lost" because having experienced the events and of subjects censored, and this is also the reason of the widening on the confidence interval over time that reflect the increasing uncertainty around the estimate of the survival function

Analysis of Addicts data in R

The `survfit2` function could also be applied to a `Surv` object, requiring the survival function to be estimated by different groups identified by a specific variable to compare $S(t)$ over groups



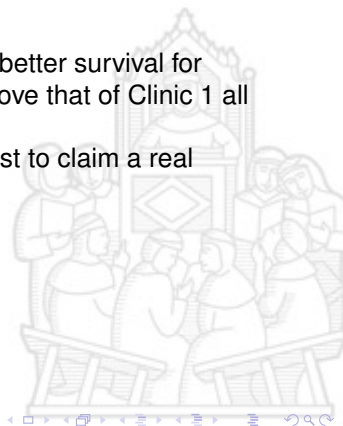
	1			
At Risk	163	87	30	1
Events	0	60	99	122
	2			
At Risk	75	50	27	7
Events	0	19	25	28



Analysis of Addicts data in R

The survival curves plotted seem to suggest a better survival for subjects treated in Clinic 2 as the curve lies above that of Clinic 1 all over the follow-up time

However, we need to do a formal hypothesis test to claim a real difference between the curves



Analysis of Addicts data in R

It is possible to test for the difference between (curves of) survival function using the `survdif` function, which by default performs the Log-Rank test

```
survdif(Surv(survt, status) ~ clinic, data = addicts)
```

Call:

```
survdif(formula = Surv(survt, status) ~ clinic, data = addicts)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
clinic=1	163	122	90.9	10.6	27.9
clinic=2	75	28	59.1	16.4	27.9

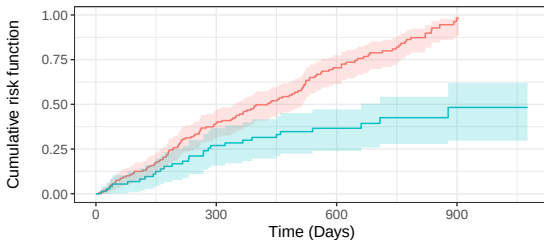
chisq= 27.9 on 1 degrees of freedom, p= 1e-07

Analysis of Addicts data in R

The test compare observed and expected events over the two groups along the follow-up time assigning equal weights to all the possible timepoints as we described in the first lecture, the P-value obtained that is quite equal to zero ($1e - 7$) allows concluding that there is a statistically significant difference between curves, and in particular subjects in Clinic 2 exhibit a better survival as compared to those in Clinic 1; that could be interpreted in terms of a greater effectiveness of Clinic 2 in retaining subjects on treatment and that may be attributed to different factors that we are not able to evaluate, such as possible better approaches used in Clinic 2 or even different characteristics of subjects that could impact on treatment retention

Analysis of Addicts data in R

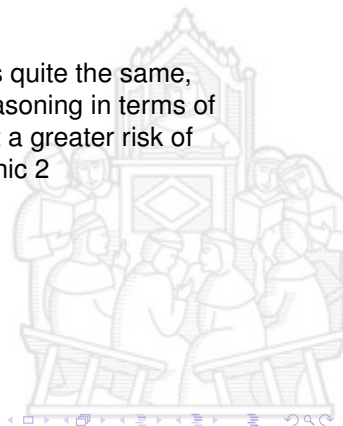
We could also have a plot of the hazard function to read data in terms of risk of treatment failure, asking the `ggsurvfit` to plot the "risk" instead of the "survival" function



1				
At Risk	163	87	30	1
Events	0	60	99	122
2				
At Risk	75	50	27	7
Events	0	19	25	28

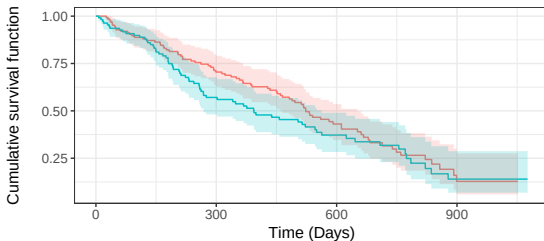
Analysis of Addicts data in R

Considering risk, interpretation of data remains quite the same, unless in this case we can interpret the plot reasoning in terms of hazard, and say that the plot seems to suggest a greater risk of treatment failure in Clinic 1 as compared to Clinic 2



Analysis of Addicts data in R

Similarly to what we have done with respect to subjects treated in the two clinics, we could try to use the same approach to assess if being imprisoned affects survival



0				
At Risk	127	84	33	4
Events	0	36	65	81
1				
At Risk	111	53	24	4
Events	0	43	59	69



Analysis of Addicts data in R

Using the Log-Rank test

```
survdif(Surv(survt, status) ~ prison, data = addicts)
```

Call:

```
survdif(formula = Surv(survt, status) ~ prison, data = addicts)
```

	N	Observed	Expected	(O-E) ² /E	(O-E) ² /V
prison=0	127	81	87.8	0.519	1.26
prison=1	111	69	62.2	0.732	1.26

Chisq= 1.3 on 1 degrees of freedom, p= 0.3

In this case, both looking at Kaplan-Meier curves, which are quite overlapping, and at the result of the Log-Rank test with P-value=0.3, suggest no difference in the survival function according to history of imprisonment